NumPy — Creating Arrays

Level 3 — Easy Examples

Each example is short and beginner-friendly. Run the code in Python after importing NumPy.

1. np.array()

Creates an array from a Python list.

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print(arr)
```

Output: [1 2 3 4]

2. np.zeros()

Creates an array filled with zeros.

```
arr = np.zeros(4)
print(arr)
```

Output: [0. 0. 0. 0.]

3. np.ones()

Creates an array filled with ones.

```
arr = np.ones(4)
print(arr)
```

Output: [1. 1. 1. 1.]

4. np.empty()

Creates an array without initializing its values. Values may look random.

```
arr = np.empty(3)
print(arr)
```

Output: Values are not fixed; they depend on memory.

5. np.full()

Creates an array filled with a chosen value.

```
arr = np.full(4, 7)
print(arr)
```

Output: [7 7 7 7]

6. np.eye()

Creates an identity-like 2D matrix with 1s on the main diagonal.

```
arr = np.eye(3)
print(arr)
```

Output: [[1. 0. 0.]
[0. 1. 0.]

[0. 0. 1.]]

7. np.identity()

Creates a square identity matrix.

```
arr = np.identity(3)
print(arr)
```

Output: [[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]]

8. np.arange()

Creates numbers with a fixed step, stopping before the end value.

```
arr = np.arange(1, 6)
print(arr)
```

Output: [1 2 3 4 5]

9. np.linspace()

Creates a chosen number of equally spaced values.

```
arr = np.linspace(0, 10, 5)
print(arr)
```

Output: [0. 2.5 5. 7.5 10.]

10. np.logspace()

Creates values spaced evenly on a logarithmic scale.

```
arr = np.logspace(1, 3, 3)
print(arr)
```

Output: [10. 100. 1000.]

11. np.geomspace()

Creates values with an equal multiplication ratio.

```
arr = np.geomspace(1, 100, 3)
print(arr)
```

Output: [1. 10. 100.]

12. np.zeros_like()

Creates zeros with the same shape as another array.

```
a = np.array([5, 6, 7])
arr = np.zeros_like(a)
print(arr)
```

Output: [0 0 0]

13. np.ones_like()

Creates ones with the same shape as another array.

```
a = np.array([5, 6, 7])
arr = np.ones_like(a)
print(arr)
```

Output: [1 1 1]

14. np.full_like()

Creates an array with the same shape as another array and a chosen value.

```
a = np.array([5, 6, 7])
arr = np.full_like(a, 9)
print(arr)
```

Output: [9 9 9]

15. np.empty_like()

Creates an uninitialized array with the same shape as another array.

```
a = np.array([5, 6, 7])
arr = np.empty_like(a)
print(arr)
```

Output: Values are not fixed; they depend on memory.